FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 4

PHYSICS / APPLIED PHYSICS

ELECTRODYNAMICS II

Module 4: Transient Circuits and Alternating Currents

Maximum Time: 2 Hours		**Maximum Marks: 70**
Section A		
All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)		
1	Show graphically the relation between alternating current and e.m.f. in a circuit containing a pure inductance. Explain the phase relationship between them.	**3**
2	Distinguish between inductive reactance and capacitive reactance. How does each vary with the frequency of the alternating current?	**3**
3	What is the j operator? Explain its geometrical significance and give one of its applications in the study of A.C. circuits.	**3**
4	Using the j operator, write down the expressions for complex inductive reactance, complex capacitive reactance, and the total complex impedance of a series LCR circuit.	**3**
5	Write down the expressions for the impedance and phase angle in a series CR circuit connected to an alternating e.m.f. Does the current lead or lag behind the applied voltage?	**3**
6	Why is a parallel resonant LC circuit called a rejector circuit? Write down the expression for its resonant frequency when the resistance of the coil is negligible.	**3**
7	Define the quality factor (Q-factor) of a series LCR circuit and give its expression. How is the Q-factor related to the bandwidth and sharpness of resonance of the circuit?	**3**
8	What is the phase difference between the current and the voltage across the resistance in a series LCR circuit at resonance? Explain voltage magnification and why the circuit is called an acceptor circuit.	**3**
9	What do you mean by wattless current? Distinguish between true power, apparent power, and power factor in an A.C. circuit.	**3**
10	Explain why there is no dissipation of average power when an alternating e.m.f. is applied to a purely inductive or a purely capacitive circuit.	**3**
Section B		
All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)		
11	An alternating sinusoidal voltage $E = E_0 \sin \omega t$ is applied separately across (i) a pure inductance L and (ii) a pure capacitance C. Derive the expressions for the instantaneous current in each case and draw the corresponding phasor diagrams.	**6**
12	Define impedance, reactance, and admittance in an A.C. circuit. A pure capacitor of $25\ \mu\text{F}$ is connected across a $220\ \text{V}$, $50\ \text{Hz}$ A.C. supply. Find the capacitive reactance, the r.m.s. value of the current, and the peak value of the current.	**6**
13	Using the j operator method, derive the expressions for the complex impedance, peak current, and phase angle when an alternating e.m.f. $E = E_0 e^{j\omega t}$ is applied to (i) a series LR circuit and (ii) a series CR circuit.	**6**
14	Explain how alternating e.m.f. and current are represented in complex form using the j operator. Apply the j operator method to obtain the complex current, actual instantaneous current, and equivalent impedance of a series LCR circuit.	**6**
15	An alternating voltage of $100\ \text{V}$ at a frequency of $25\ \text{Hz}$ is applied to a circuit consisting of a resistance $1.5\ \Omega$ and an inductance of $0.01\ \text{H}$ in series: (a) find the current flowing; (b) phase difference between e.m.f. and current; and (c) potential drop across resistor and inductor.	**6**
16	An alternating potential of $100\ \text{V}$ and $50\ \text{Hz}$ is applied across a series circuit having an inductance of $5\ \text{H}$, a resistance of $100\ \Omega$, and a variable capacitance. At what value of capacitance will the current in the circuit be in phase with the applied voltage? Calculate the current and the potential differences across the resistance, inductance, and capacitance in this condition.	**6**
17	Derive an expression for the average power consumed over a complete cycle in an A.C. circuit. An alternating voltage of $10\ \text{V}$ at $100\ \text{Hz}$ is applied to a choke of inductance $5\ \text{H}$ and resistance $200\ \Omega$. Determine the power factor of the coil and the power absorbed.	**6**

18	A circuit consists of a non-inductive resistance of $50\ \Omega$, an inductance of $0.3\ \text{H}$ and a resistance of $2\ \Omega$, and a capacitor of $40\ \mu\text{F}$ in series and is supplied with $200\ \text{V}$ at $50\ \text{Hz}$. Find the impedance, the current, lag or lead, the power factor, and the power in the circuit.	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	Obtain expressions for the resultant e.m.f., impedance, and power factor of an LCR series circuit when an alternating current is flowing through it. Explain the resonance, sharpness of resonance, bandwidth, and Q-factor in an LCR series circuit.	**10**
20	Investigate the behaviour of a parallel resonant circuit consisting of a capacitance C in parallel with a coil of inductance L and resistance R connected across an A.C. source. Derive expressions for its resonant frequency and impedance at resonance, and compare series and parallel resonant circuits.	**10**